

LFR (Line Following Robot)

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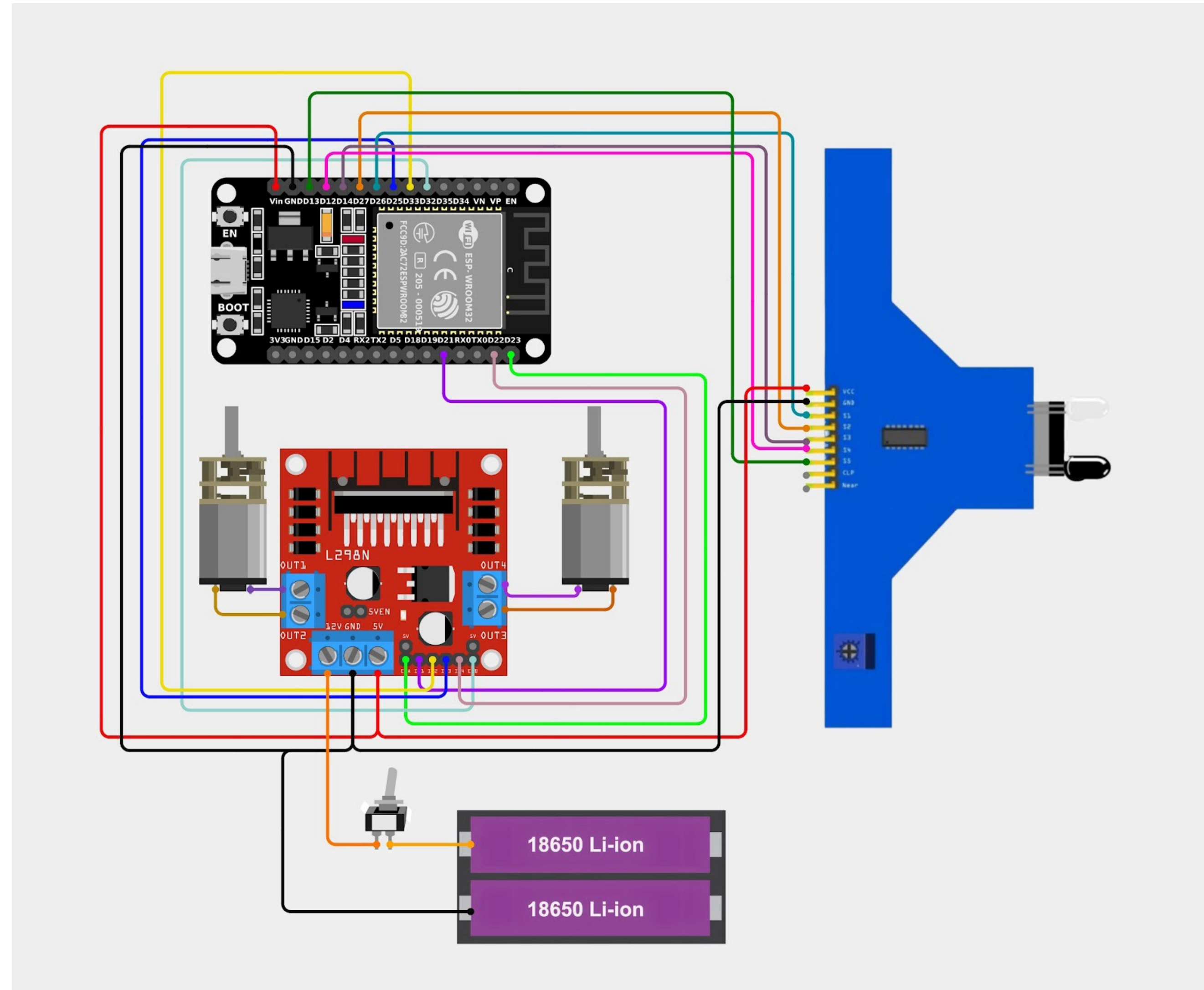
General Secretary at Programming and AI Club.



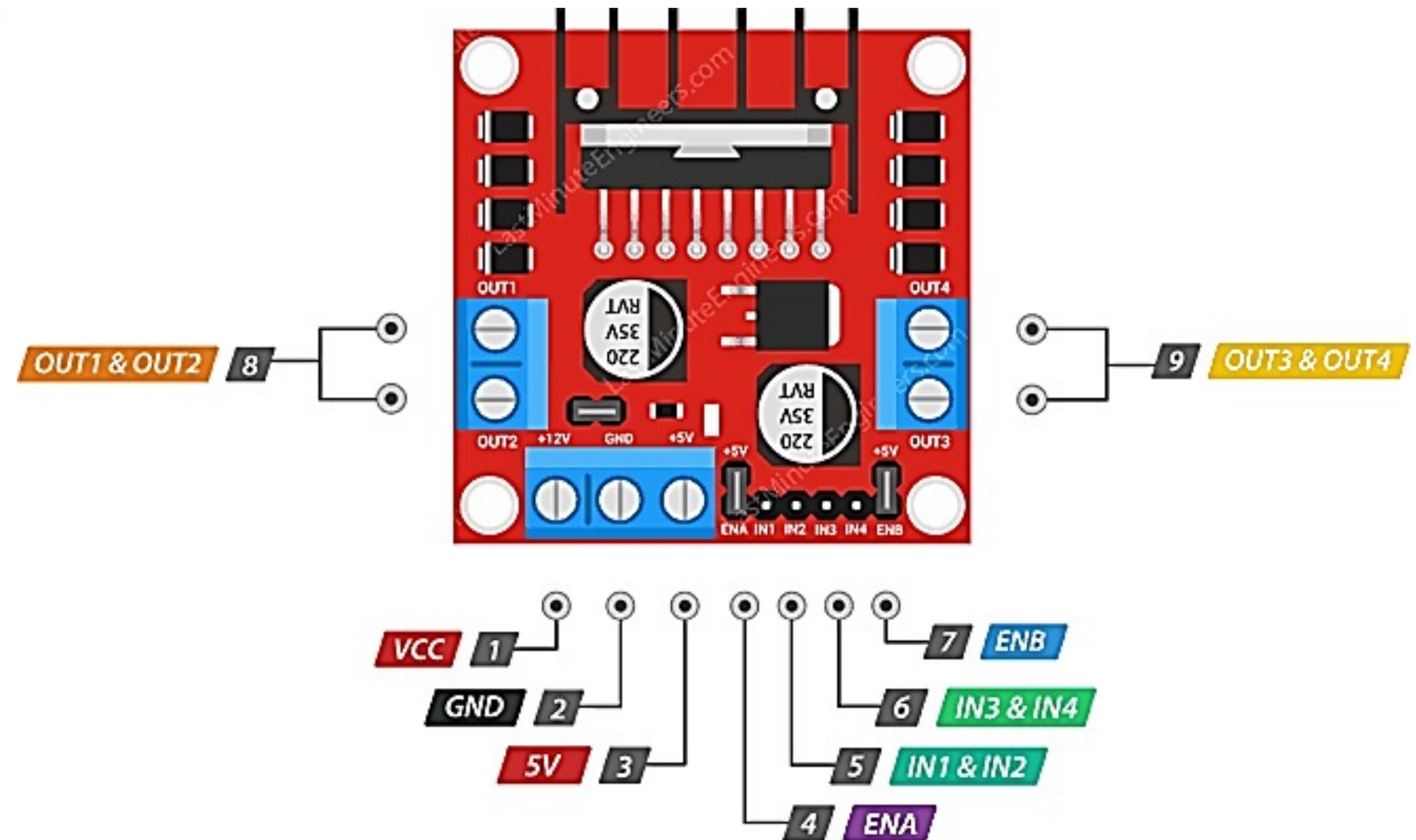
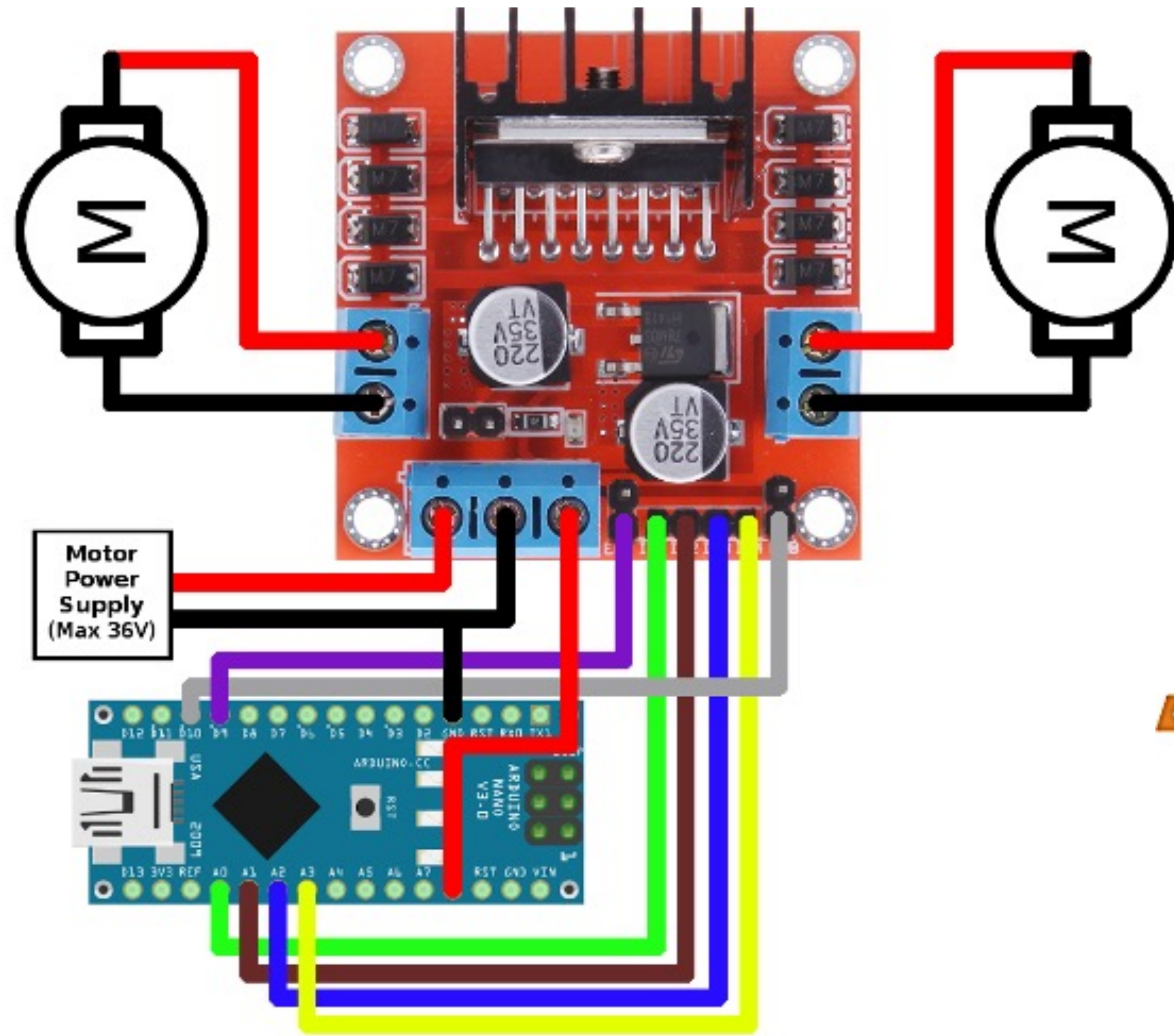
Component List

Microcontroller	Arduino: mega, nano, Uno or STM32 Series or ESP32 Series
IR sensor	Analog or Digital , 5 or 6 or 8 Array IR sensor . Or Custom made multiple Array IR sensor.
Motor ,Motor mount and Wheel,Ball Caster	12V N20 Gear Motor(100-200 RPM for beginner) or 16GA-050 12V DC Gear Motor
Motor Driver	L298N H-Bridge Dual Motor Driver or Motor Driver - Dual TB6612FNG (1A)
Chassis	Veroboard Dot Type or 3D Printed Chassis or PCB
Battery	12V Lipo Battery
Buck Step-down Converter	5V Fixed Output DC-DC Buck Step-down Converter. Or HW-411A LM2596 DC To Dc Buck Converter Step Down Module Power Supply
Others Component	Push Button 4 pin, I2C OLED Display, LED.

LFR Circuit



Motor Driver Connection



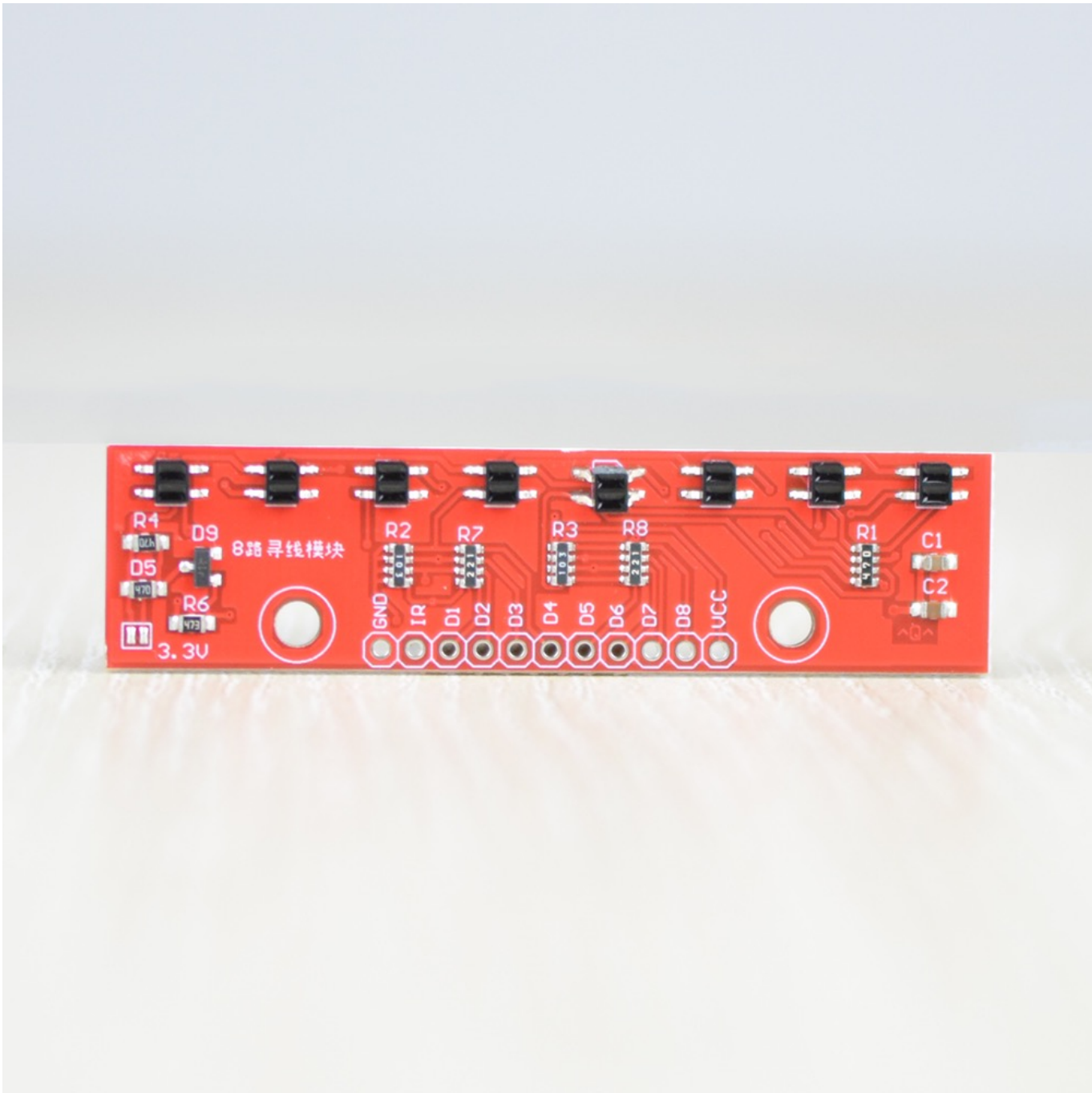
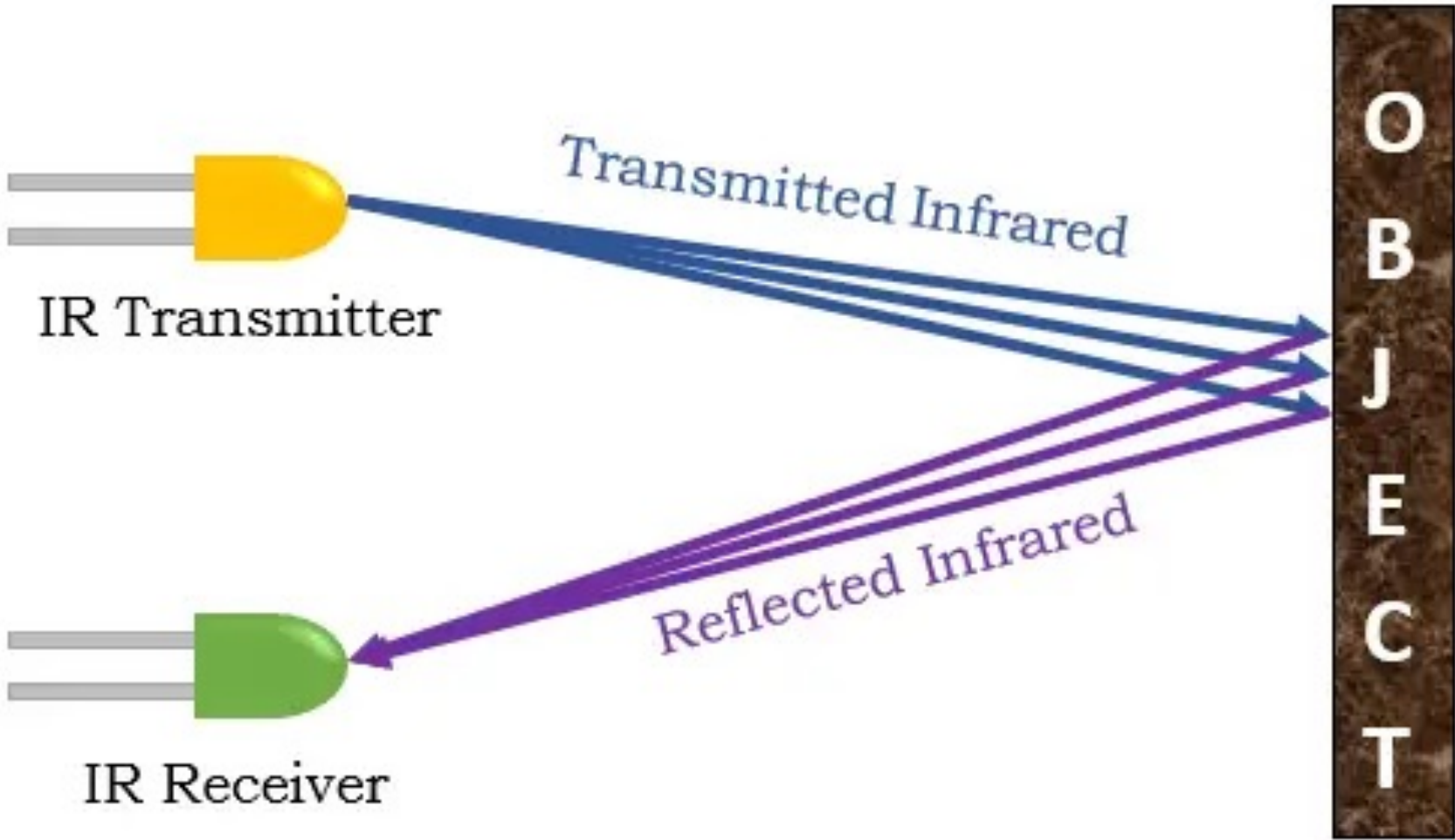
PWM PIN

PWM stands for Pulse Width Modulation.
A PWM pin is a digital output pin that can simulate an analog voltage by turning on and off very quickly.

UNO or nano	3, 5, 6, 9, 10, 11
Mega or Mega mini	2 to 13, 44, 45, 46
32 ESP	Almost all GPIOs (0-33), except GPIOs 34-39



IR - Sensor

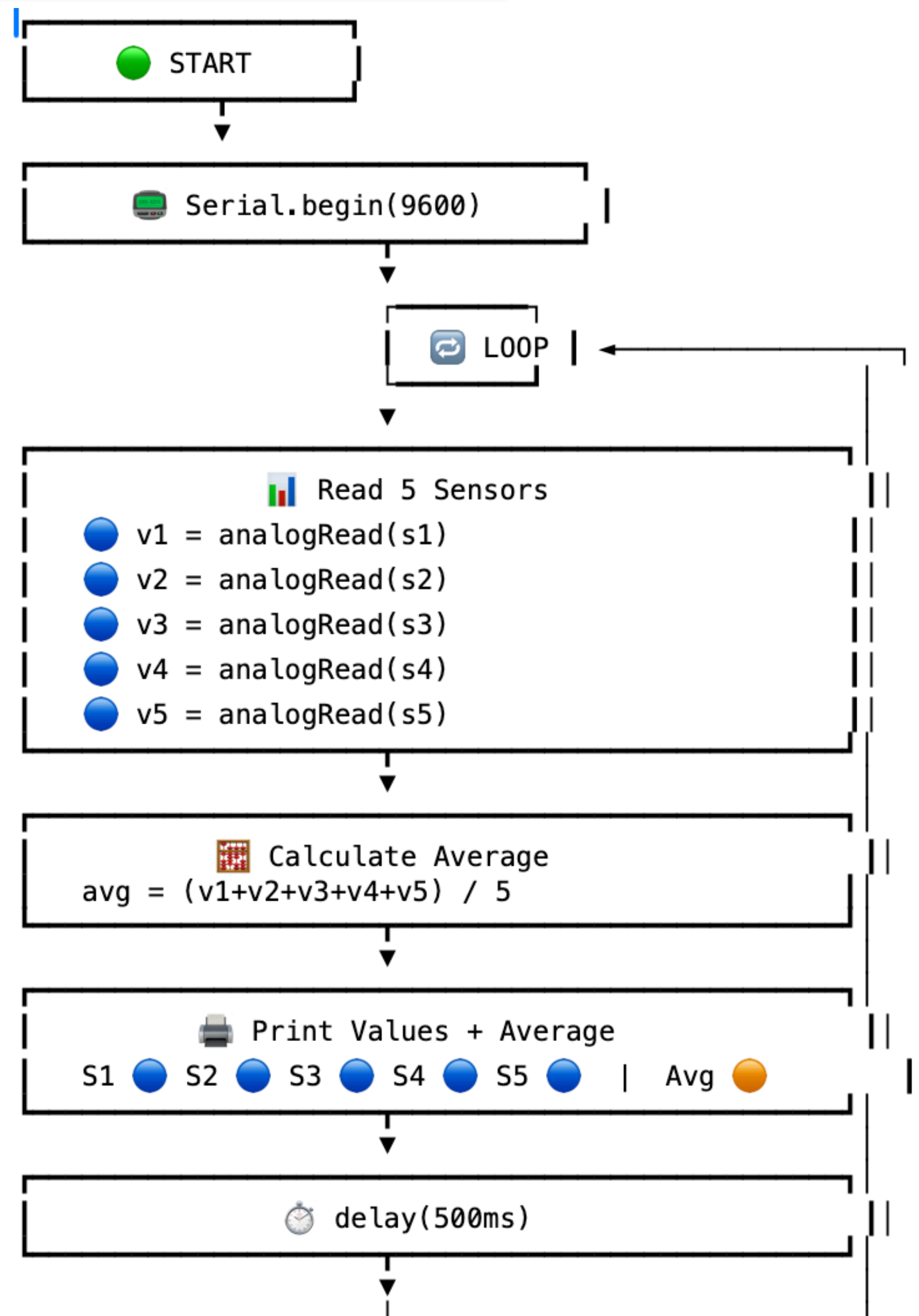


Calibration

// Sensor Calibration & Threshold Finder

```
int s1 = A0;  
int s2 = A1;  
int s3 = A2;  
int s4 = A3;  
int s5 = A4;
```

```
void setup() {  
  Serial.begin(9600);  
  delay(1000);  
}
```



```
void loop() {  
  // Read sensors  
  int v1 = analogRead(s1);  
  int v2 = analogRead(s2);  
  int v3 = analogRead(s3);  
  int v4 = analogRead(s4);  
  int v5 = analogRead(s5);
```

```
  // Calculate average  
  int avg = (v1 + v2 + v3 + v4 + v5) / 5;
```

```
  // Display values  
  Serial.print("S1:");  
  Serial.print(v1);  
  Serial.print(" S2:");  
  Serial.print(v2);  
  Serial.print(" S3:");  
  Serial.print(v3);  
  Serial.print(" S4:");  
  Serial.print(v4);  
  Serial.print(" S5:");  
  Serial.print(v5);  
  Serial.print(" | Avg:");  
  Serial.println(avg);
```

```
  delay(500);  
}
```

Coding Part

Function	What it does
setup()	Sets pins as INPUT/OUTPUT (runs once)
loop()	Reads sensors and makes decisions (runs forever)
forward()	Robot moves straight
left()	Robot turns left (gentle)
right()	Robot turns right (gentle)
hardLeft()	Robot turns left sharply (hard turn)
hardRight()	Robot turns right sharply (hard turn)
stop()	Robot stops

Define Pin and Sensor threshold

```
// Sensor pins
int s1 = A0;    // Leftmost sensor
int s2 = A1;    // Left sensor
int s3 = A2;    // Center sensor
int s4 = A3;    // Right sensor
int s5 = A4;    // Rightmost sensor

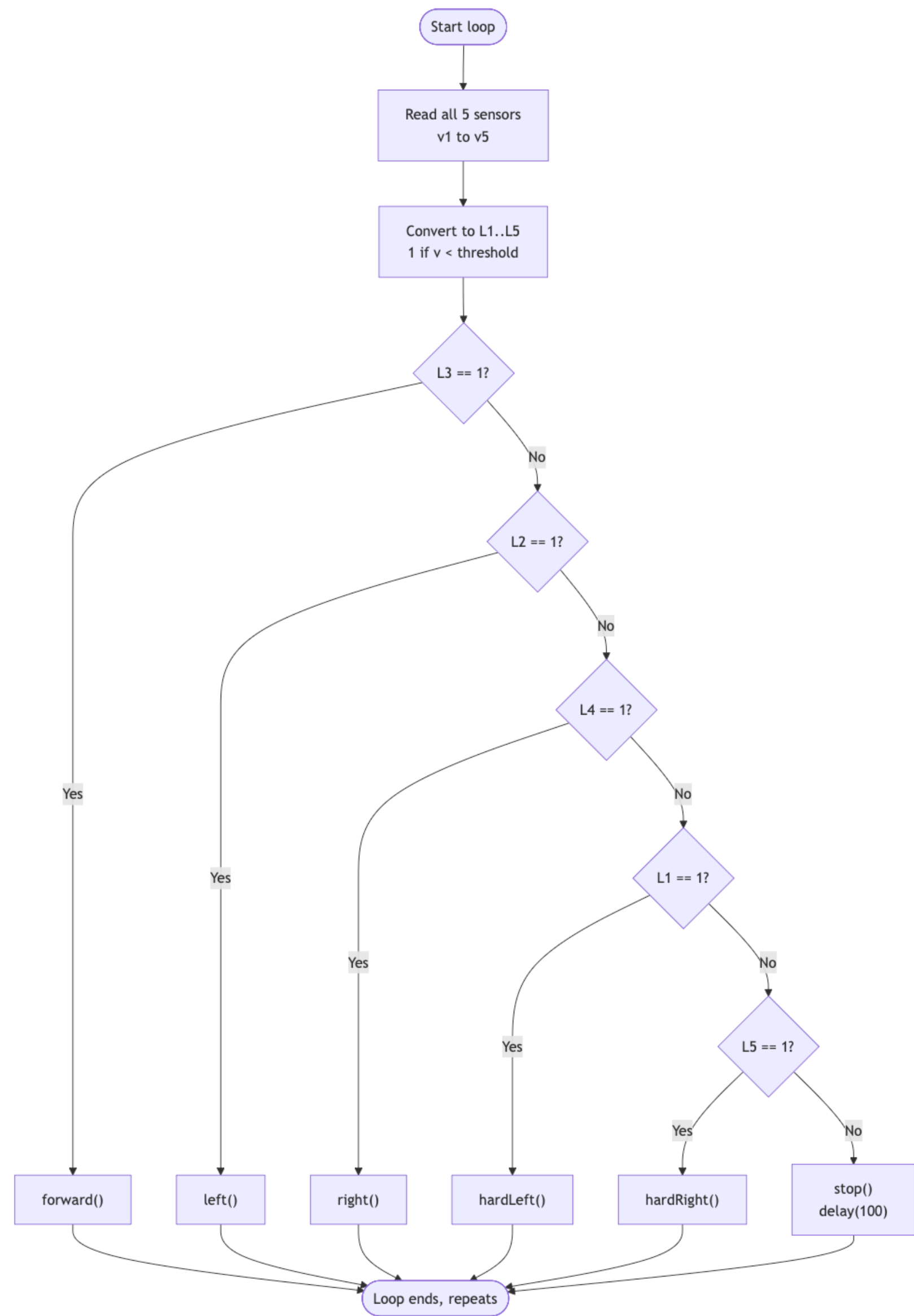
// Motor pins
int lm1 = 7;    // Left motor pin 1
int lm2 = 8;    // Left motor pin 2
int lpwm = 5;   // Left motor speed (PWM)
int rm1 = 9;    // Right motor pin 1
int rm2 = 10;   // Right motor pin 2
int rpwm = 6;   // Right motor speed (PWM)

int th = 500;   // Sensor threshold (below this = black line)
```


setup Function

```
void setup() {  
    // Set sensor pins as input  
    pinMode(s1, INPUT);  
    pinMode(s2, INPUT);  
    pinMode(s3, INPUT);  
    pinMode(s4, INPUT);  
    pinMode(s5, INPUT);  
  
    // Set motor pins as output  
    pinMode(lm1, OUTPUT);  
    pinMode(lm2, OUTPUT);  
    pinMode(rm1, OUTPUT);  
    pinMode(rm2, OUTPUT);  
    pinMode(lpwm, OUTPUT);  
    pinMode(rpwm, OUTPUT);  
}
```

loop Function



```
void loop() {  
  // Read all 5 sensors  
  int v1 = analogRead(s1);  
  int v2 = analogRead(s2);  
  int v3 = analogRead(s3);  
  int v4 = analogRead(s4);  
  int v5 = analogRead(s5);
```

```
  // Detect black line (sensor value below threshold)  
  int L1 = (v1 < th) ? 1 : 0; // 1 = line detected  
  int L2 = (v2 < th) ? 1 : 0;  
  int L3 = (v3 < th) ? 1 : 0;  
  int L4 = (v4 < th) ? 1 : 0;  
  int L5 = (v5 < th) ? 1 : 0;
```

```
  // Line following logic based on which sensor sees the line  
  if (L3 == 1) {  
    // Center sensor on line → go straight  
    forward();  
  }  
  else if (L2 == 1) {  
    // Left sensor on line → turn left  
    left();  
  }  
  else if (L4 == 1) {  
    // Right sensor on line → turn right  
    right();  
  }  
  else if (L1 == 1) {  
    // Leftmost sensor on line → hard left turn  
    hardLeft();  
  }  
  else if (L5 == 1) {  
    // Rightmost sensor on line → hard right turn  
    hardRight();  
  }  
  else {  
    // No sensor detects line → stop and wait  
    stop();  
    delay(100);  
  }  
}
```


Move forward

```
// Move forward  
void forward() {  
    digitalWrite(lm1, HIGH);  
    digitalWrite(lm2, LOW);  
    digitalWrite(rm1, HIGH);  
    digitalWrite(rm2, LOW);  
    analogWrite(lpwm, 150);  
    analogWrite(rpwm, 150);  
}
```

Turn left (gentle)

```
// Turn left (gentle)
void left() {
    digitalWrite(lm1, LOW);
    digitalWrite(lm2, HIGH);
    digitalWrite(rm1, HIGH);
    digitalWrite(rm2, LOW);
    analogWrite(lpwm, 100);
    analogWrite(rpwm, 100);
}
```

Turn right (gentle)

```
// Turn right (gentle)
void right() {
    digitalWrite(lm1, HIGH);
    digitalWrite(lm2, LOW);
    digitalWrite(rm1, LOW);
    digitalWrite(rm2, HIGH);
    analogWrite(lpwm, 100);
    analogWrite(rpwm, 100);
}
```


Hard left turn

```
// Hard left turn (sharp)
void hardLeft() {
    digitalWrite(lm1, LOW);
    digitalWrite(lm2, HIGH);
    digitalWrite(rm1, HIGH);
    digitalWrite(rm2, LOW);
    analogWrite(lpwm, 120);
    analogWrite(rpwm, 120);
}
```

Hard right turn

```
// Hard right turn (sharp)
void hardRight() {
    digitalWrite(lm1, HIGH);
    digitalWrite(lm2, LOW);
    digitalWrite(rm1, LOW);
    digitalWrite(rm2, HIGH);
    analogWrite(lpwm, 120);
    analogWrite(rpwm, 120);
}
```